

WAGA SKYVIEW DISPLAY FLASHING INSTRUCTIONS

Use these instructions to quick start a new LilyGo T5S board or load a new version of the software.

If loading a new version of software, it is recommended to take a photograph of the WebGUI settings menu before flashing the new version so that you can reapply your personalised settings.

1. Download the 'Flash Download Tool' from www.espressif.com.



The screenshot shows a web browser window with the URL https://docs.espressif.com/projects/esp-test-tools/en/latest/esp32/production_stage/tools/flash_download_tool.html. The page title is 'Flash Download Tool User Guide'. On the left, there is a sidebar with a search bar and a table of contents. The table of contents has two main sections: 'DEVELOPMENT STAGE' and 'PRODUCTION STAGE'. Under 'DEVELOPMENT STAGE', there are links for 'Introduction', 'RF Test Guide', 'RF Test Items', 'RF Test Certification', and 'WFA Certification and Test Guide'. Under 'PRODUCTION STAGE', there is a link for 'Introduction'. Below the table of contents, there are links for 'Flash Download Tool', 'Preparation', 'Tool Overview', and 'Download Example'. The main content area on the right has a title 'Flash Download Tool User Guide' with a '[中文]' link. Below the title is a section titled 'Preparation' with the text 'The software and hardware resources required for downloading firmware to flash are listed below.' followed by a bulleted list: 'Hardware:' with sub-points '1 x module to which firmware is downloaded' and '1 x PC (Windows 7 [64 bits], Windows 10)'; and 'Software:' with a sub-point 'Flash Download Tool' with a download icon.

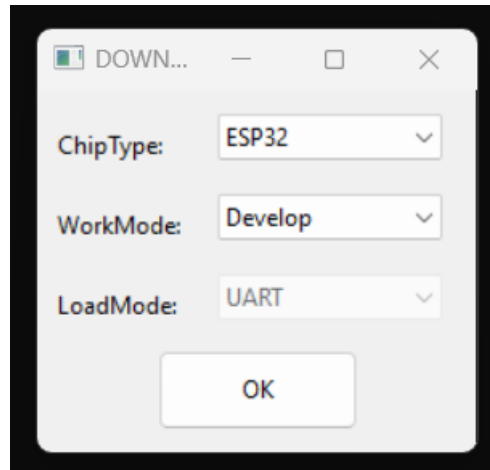
https://docs.espressif.com/projects/esp-test-tools/en/latest/esp32/production_stage/tools/flash_download_tool.html

It will download as a zip file, so extract the files. It will look something like this:

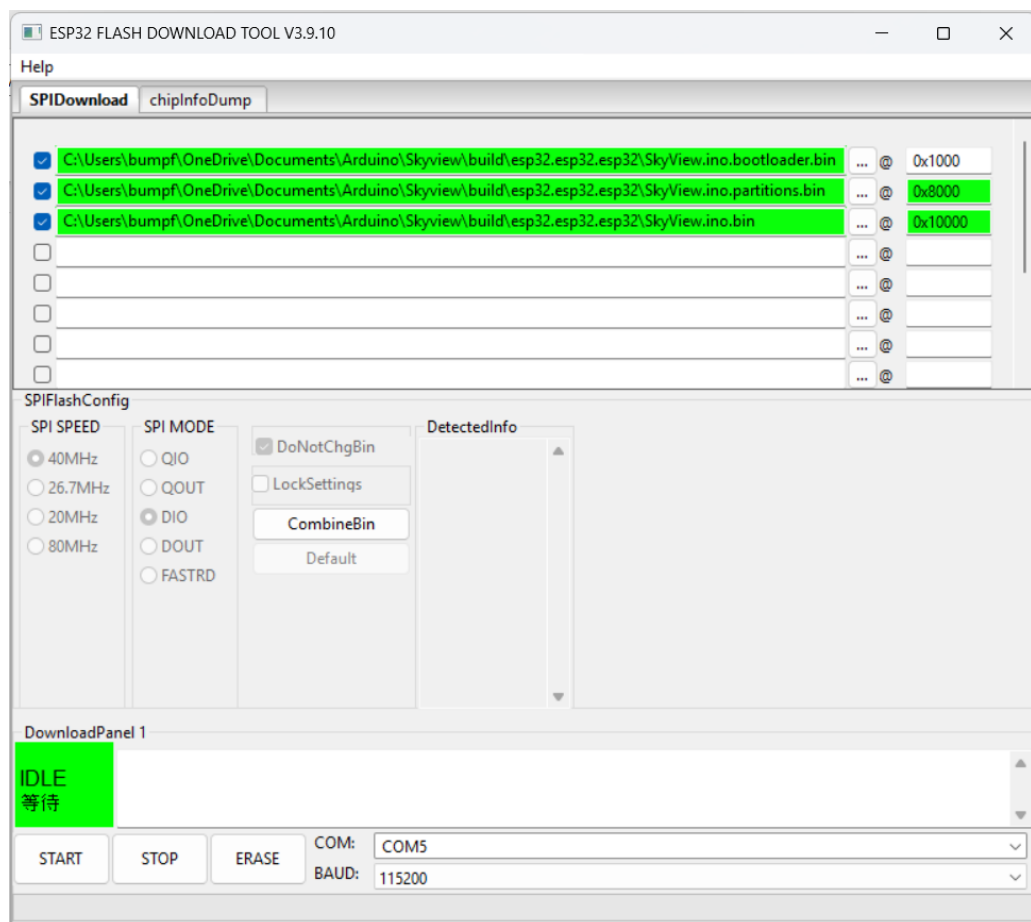
 flash_download_tool_3.9.10	16/04/2026 8:11 PM	Application
 flash_dump	16/04/2026 8:58 PM	File folder
 combine	16/04/2026 8:20 PM	File folder
 logs	16/04/2026 8:18 PM	File folder
 dl_temp	16/04/2026 8:16 PM	File folder
 configure	16/04/2026 8:11 PM	File folder

2. Connect the LilyGo T5S board to a PC using a USB cable with USB micro plug for the T5S board. This will power the board and enable data to be transferred. Remove any micro-SD card that is installed.
3. Find the WAGA SkyView Display binaries at: https://github.com/Bumpff/SoftRF_Skyview-WAGA and copy them to a directory of your choice on your PC. There should be 3 files.
...ino.binloader.bin
...ino.partition.bin
...ino.bin

4. Start the 'flash_download_tool...' application. You will get this screen and must select ESP32 and Develop options. Then press 'OK'.



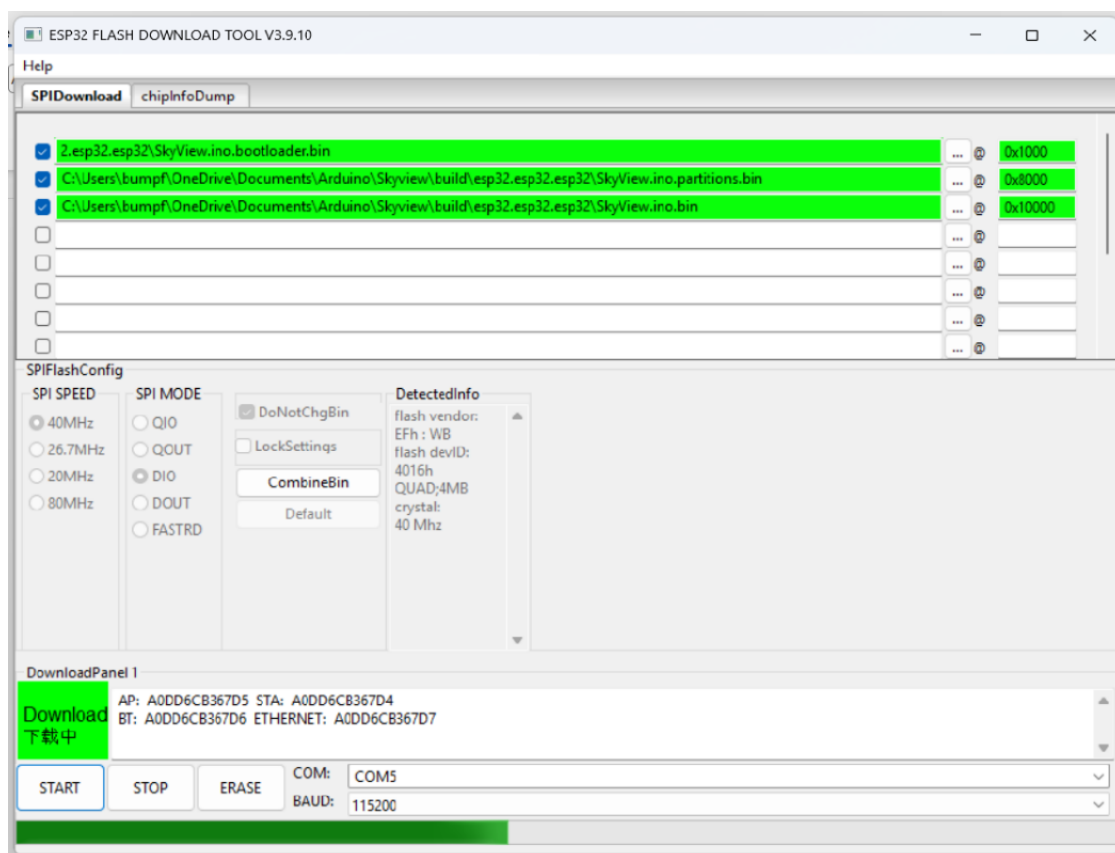
5. An application screen will open, with 2 tabs. Select the SPIDownload tab. The screen has a table for files at the top, SPIFlashConfig settings in the middle and DownloadPanel 1 at the bottom.



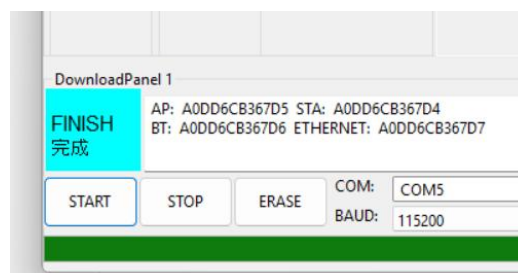
With tool version V3.9.10, it automatically sets the required SPIFlashConfig settings for the ESP32 processor.

6. At bottom right, select the COM number of the PC com port that is being used eg COM5. Set the baud rate to match the PC. Use Windows Device Manager to view and set PC Port properties. (Baud = 115,200 with data/parity/stop set to 8N1, works). A mismatch of baud rate is the most common cause of the flashing not starting.

7. In the file table, load the 3x files for flashing by pressing the ellipsis button (...) to open the file explorer and select one file at a time. As it loads, the file name will appear in the table and it will be highlighted green when loading is completed. Repeat for the other flash files.
8. Assign memory addresses 0x1000, 0x8000 and 0x10000 to the specific flash file types shown in the picture. Do not deviation from these addresses.
9. Ensure the left-hand column is ticked which will highlight the file name in green. The table can included multiple files but only the highlighted files will be loaded.
10. Press 'ERASE' to clear the board's memory. This is not mandatory but is advised.
11. Press 'START'. The loading will begin, the status indicator will show 'Download' and a timer bar shows the progress.



12. When it is complete the status indicator will change to 'FINISH' with a blue background.



13. Reboot the SkyView using the Reset button and it should come to life using the default first startup configuration settings. The default first start-up configuration settings are shown below.

3:32 No internet connection

192.168.1.1/settings

Settings

Display adapter	e-Paper TTGO T5S
Connection type	Serial
Protocol	NMEA
Baud rate	19200
Source Id	SoftRF-abc123
Key	12345678
Bridge Output	None
Units	mixed
Screen layout	Portrait
View mode	Radar
Radar orientation	Track Up
Zoom level	low
Nav Waypoint	YBEV
Aircrafts data	GliderNet
ID preference	tail/CN
Sound	Buzzer
e-Paper 'ghosts' removal	off
Traffic advisories filter	Unfiltered
Power save	Disabled
Team Member Id	000000

Save and restart

Default settings that are applied during flashing

14. Refer to the User Manual for details on how to use a smartphone to access the WebGUI configuration menu to personalise the settings.